

Swinburne University of Technology
School of Science, Computing, and Emerging Technologies

SWE40006 - Software Deployment and Evolution
Deployment Portfolio Task 1

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Deployment Portfolio Task 1

1 Declared Target Level

All four subtasks 1.1, 1.2, 1.3, and 1.4 were attempted.

2 Execution Evidence

2.1 Task 1.1

I built and packaged a sample console application into an .msi installer using WiX 7.0.0 via the Command-line Interface (CLI).

The first step was creating a new C# project, adding the provided code, and building it in Release mode.

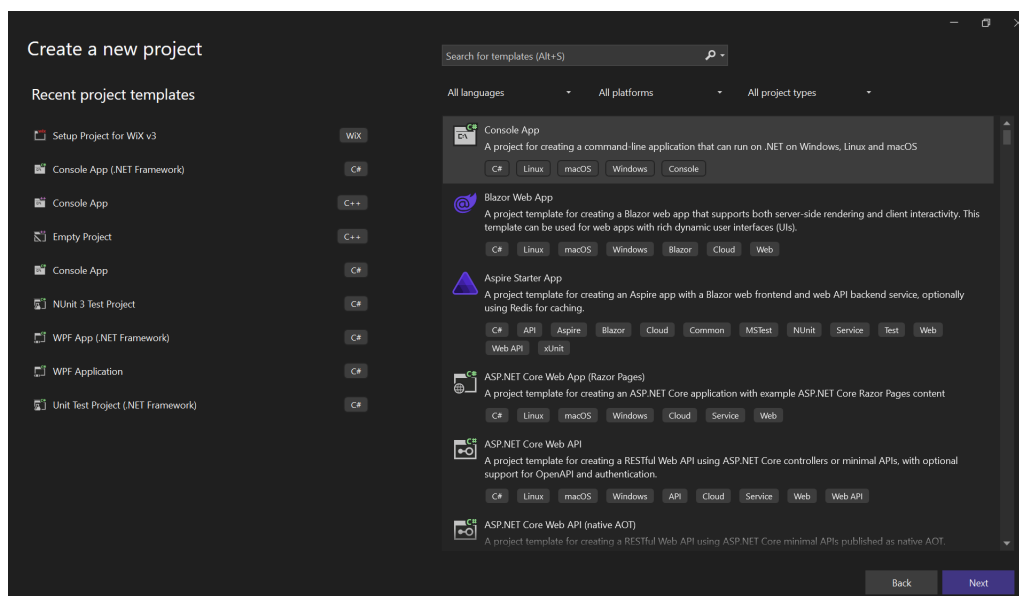


Figure 1: Creating a new C# console application.

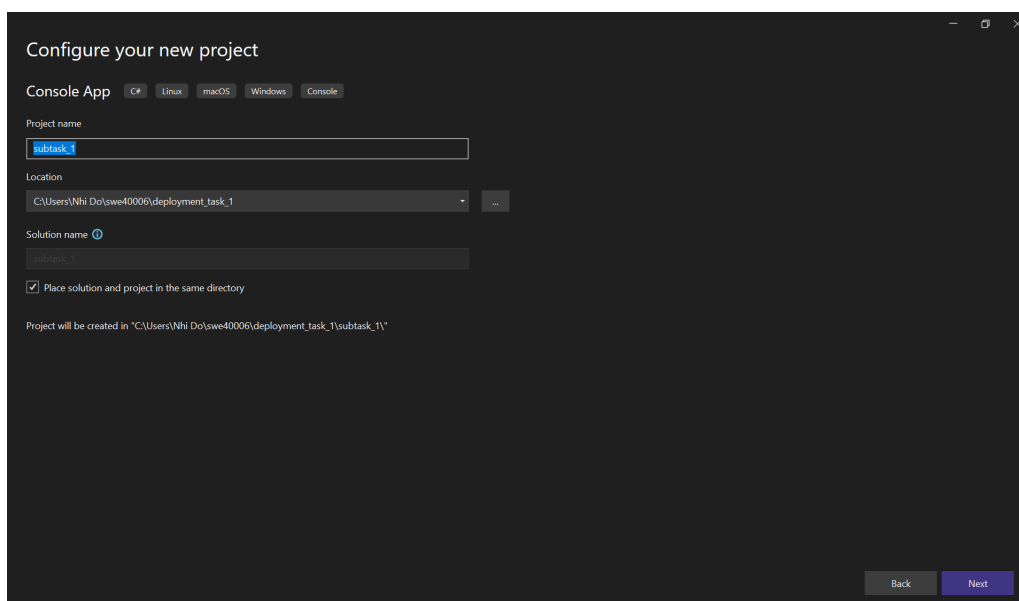


Figure 2: Naming the application.

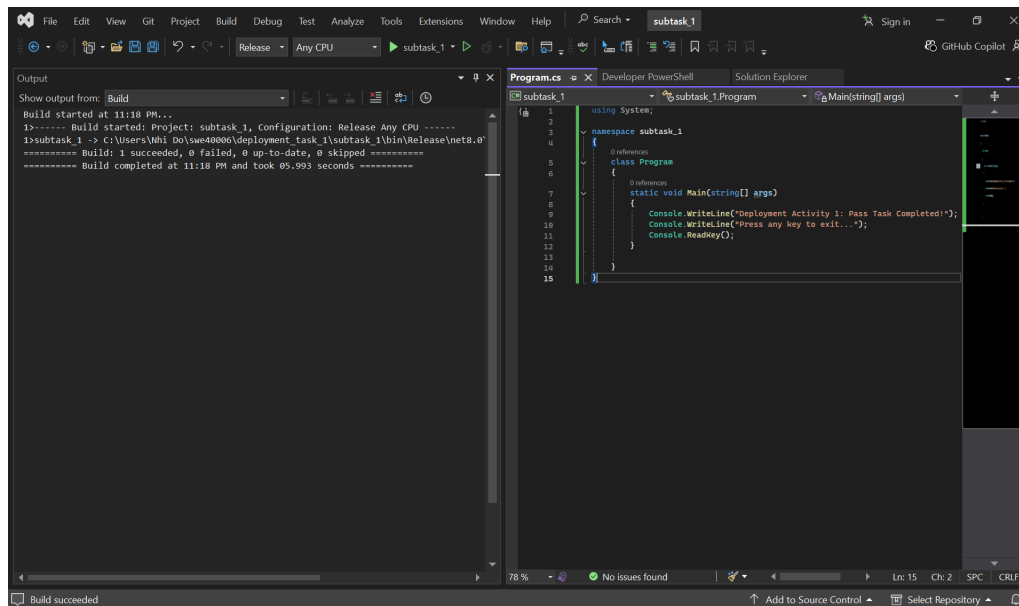


Figure 3: Building the program in Release mode.

Once the program has been built, its executable is created the project's **bin Release** folder.

The WiX toolset was installed using the CLI.

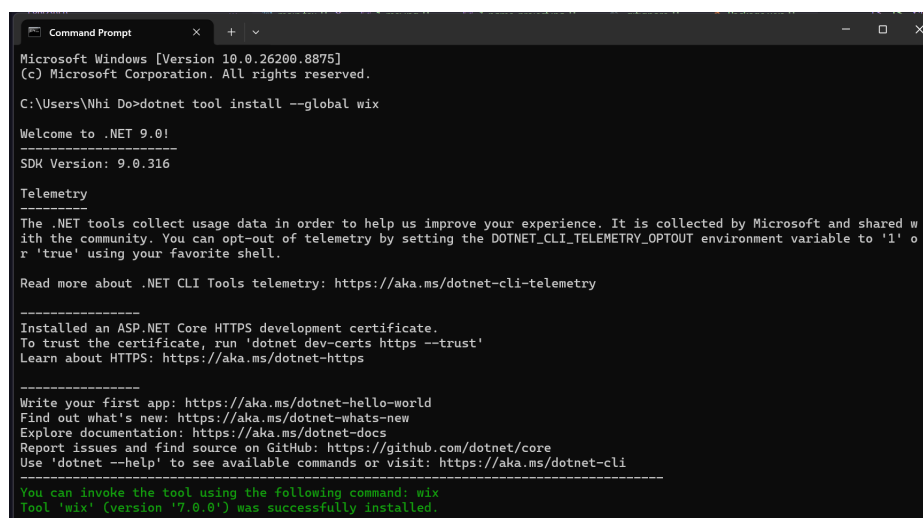


Figure 4: Installing WiX.

The WiX project and source file were manually created. The **.wxs** file was then configured to contain instructions to build the installer.

The screenshot shows the Visual Studio interface with the 'subtask_1' project selected. The 'Developer PowerShell' window is open, displaying the following commands and output:

```
PS C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1> mkdir wix_installer

Directory: C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1

Mode                LastWriteTime         Length Name
----                -
d-----            8/8/2026 11:26 PM             wix_installer

PS C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1> cd .\wix_installer\
PS C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1\wix_installer> type nul > Package.wxs
type : cannot find path 'C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1\wix_installer\nul' because it does not exist.
At line:1 char:1
+ type nul > Package.wxs
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Users\Nhi Do\...x_installer\nul:String) [Get-Content], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetContentCommand

PS C:\Users\Nhi Do\swe40006\deployment_task_1\subtask_1\wix_installer>
```

Figure 5: Creating the WiX installer project and source file.

The screenshot shows the Visual Studio interface with the 'subtask_1' project selected. The 'Package.wxs' file is open in the editor, displaying the following XML content:

```
1 <Wix xmlns="http://wixtoolset.org/schemas/vd/wxs">
2
3   <Package
4     Name="Subtask 1"
5     Manufacturer="Sainmy Student"
6     Version="1.0.0"
7     UpgradeCode="1fcb3dfc-9388-4898-b478-07c0b6923c1">
8
9     <MajorUpgrade
10       DowngradeErrorMessage="A newer version of Subtask 1 is already installed." />
11
12     <MediaTemplate />
13
14     <StandardDirectory Id="ProgramFilesFolder">
15       <Directory Id="INSTALLFOLDER" Name="subtask_1" />
16     </StandardDirectory>
17
18     <Feature Id="MainFeature" Title="subtask_1" Level="1">
19       <ComponentGroupRef Id="AppFiles" />
20     </Feature>
21
22   </Package>
23
24   <Fragment>
25     <ComponentGroup Id="AppFiles" Directory="INSTALLFOLDER">
26       <File Source="..\bin\Release\net8.0\subtask_1.exe" />
27       <File Source="..\bin\Release\net8.0\subtask_1.dll" />
28       <File Source="..\bin\Release\net8.0\subtask_1.deps.json" />
29       <File Source="..\bin\Release\net8.0\subtask_1.runtimeconfig.json" />
30     </ComponentGroup>
31   </Fragment>
32
33 </Wix>
```

Figure 6: Configuring the Package.wxs file to build the installer.

The installer is then built using the command below. The resulting MSI file is created inside the installer directory.

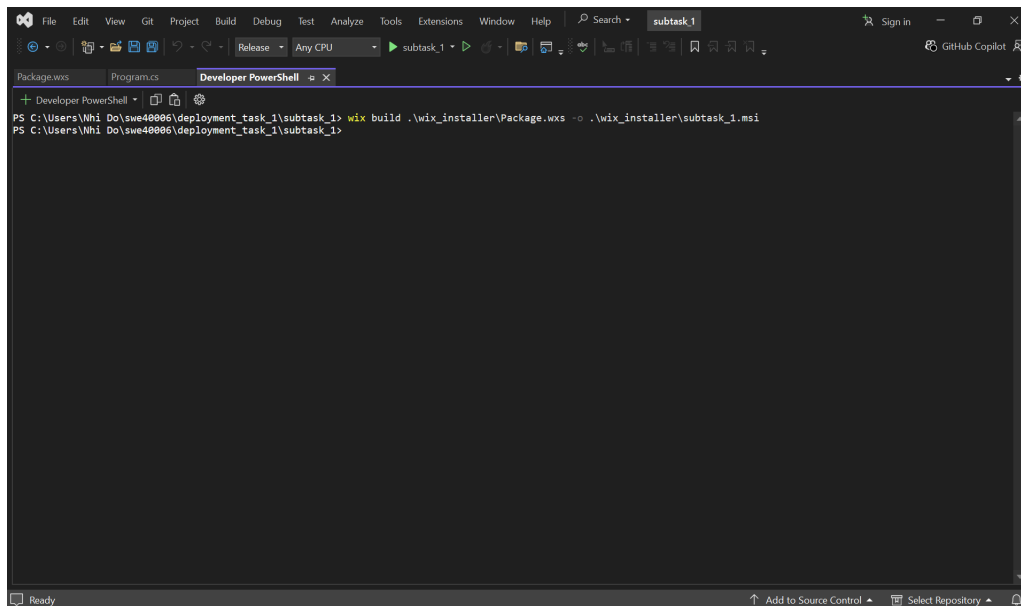


Figure 7: Building the installer from the WiX source file.

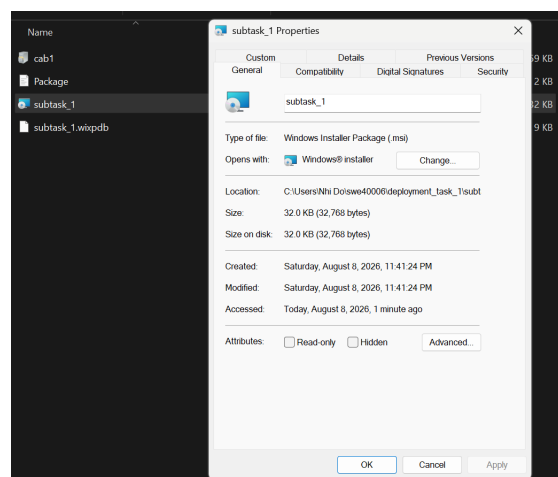


Figure 8: The MSI file is created.

Running the installer `subtask_1.msi` installs the application. The screenshot below shows that the app is successfully installed in the Program Files(x86) folder and runs as expected.

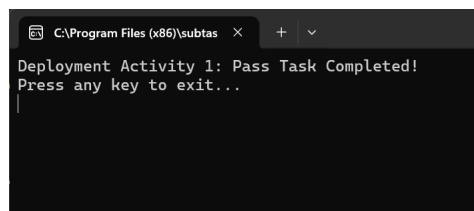


Figure 9: Running the console application after it has been installed.

The application `subtask_1` is finally uninstalled.

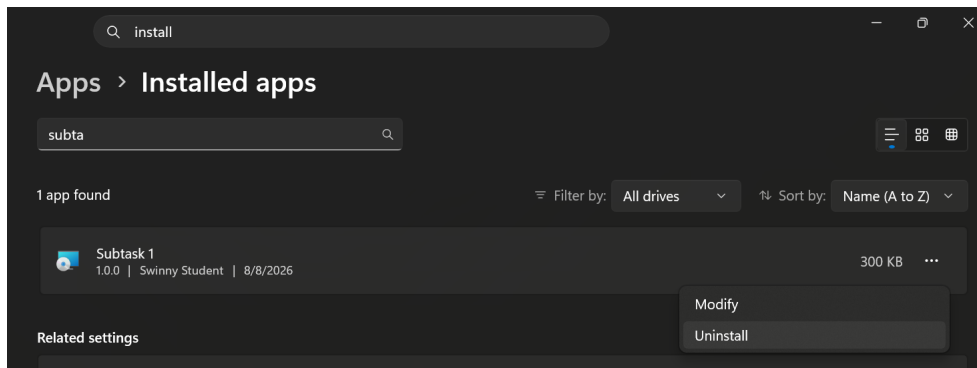


Figure 10: Uninstalling the application.